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Unconventional Suppression of Antiferromagnetic Spin Fluctuation in the High-T_c Cu Oxide $La_{2-x}Sr_xCuO_4$

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The doping dependence of low-frequency spin dynamics is discussed from a viewpoint of the self-consistently renormalized spin-fluctuation theory for the experimental planar Cu nuclear spin-lattice relaxation rate $1/T_1$ in the doped high-T_c Cu oxides, $La_{2-x}Sr_xCuO_4$ ($x = 0.06$, 0.15 and 0.20). We report that the crossover temperature T^* of T_1T deviating from the high-T Curie-Weiss law increases with reduced carrier density. We found that the doping dependence of T^* is not consistent with the current spin-fluctuation theories in which the actual metal-insulator transition is regarded as the disorder-to-order transition. The present suppression of the staggered spin fluctuation is an anomaly near half-filling, inclusive of the pseudo spin-gap.

KEYWORDS: NMR, high-T_c cuprate, spin fluctuation, pseudo spin-gap, metal-insulator transition

The anomalous magnetic properties of high-T_c Cu oxides have attracted much interest in the studies of strongly correlated metal in two spatial dimensions (2D).[1)] High-T Curie-Weiss (CW) staggered spin susceptibility $\chi(\boldsymbol{Q})$ with $\boldsymbol{Q} \sim (\pi, \pi)$ is observed over a wide doping range by the planar Cu NMR/NQR techniques,[2)] and is explained by the *thermal* antiferromagnetic spin fluctuations in the self-consistent renormalization (SCR) theory.[2,3)] Here, we call the thermal-fluctuation dominance the high-T. However, the dynamics toward the $T = 0$ magnetic-nonmagnetic transition which takes place by varying the carrier density has not been sufficiently understood. Measurement of the planar Cu nuclear spin-lattice relaxation rate $1/T_1$ provides us with information on the low-frequency spin dynamics of the CuO_2 plane,[4)] and that of the Gaussian decay rate $1/T_G$ provides us with information on the spin fluctuation spectrum integrated up to higher frequency.[5,6)] For 2D nearly antiferromagnetic metal, the expressions of $1/T_1T \sim \xi^2/(T_AT_0)$ and $(1/T_G)^2 \sim \xi^2/(T_A)^2$ are obtained in terms of the antiferromagnetic correlation length ξ ($\gg$the CuO_2 lattice constant), the spin-fluctuation energy T_0 and the spatial spread T_A.[3,7)] Up to now, the following anomalies have been revealed from measurements of the planar Cu $1/T_1$ and the Gaussian decay rate $1/T_G$.

(i) The sign of Weiss temperature Θ in the high-T CW law $1/(T+\Theta)$ of $1/T_1T$ [or $(1/T_G)^2$] changes around the optimal doping, i.e., $\Theta < 0$ in the underdoping regime and $\Theta > 0$ in the overdoping one.[2,8-10)]

(ii) The relxation rate $1/T_1T$ [or $(1/T_G)^2$] follows a CW law at high T, whereas it deviates from this law and is suppressed at low T. A crossover temperature T^* exists.[2,9-11)]

(iii) For the bilayer $YBa_2Cu_3O_y$ ($6.42 < y < 7$), suppression of $1/T_1T$ is promoted. With decreasing T, $1/T_1T$ deviates from $(1/T_G)^2$, reaches a maximum value at T_s ($=95 \sim 160$ K) and finally declines to T_c.[2,12,13)] This T-dependence of $1/T_1T$ is called the pseudo spin-gap behavior.[1)] The pseudo spin-gap behavior is recently confirmed for the monolayer.[14)]

The 2D t-J model has been a prototype model for the low-energy physics of the high-T_c Cu oxides.[15-17)] The gauge-field fluctuation is shown to have a strong pair-breaking effect on the in-plane singlet formation,[17)] whereas Kishine demonstrated for a bilayered itinerant-Heisenberg model that the interlayer dynamical screening effect suppresses the in-plane triplet formation with remaining singlet pairing at low T.[18)] His theory also accounted for the low-T departure of $1/T_1T$ from $(1/T_G)^2$ for the bilayer systems without dynamical anomaly. The experimental doping dependence of T_s has also been studied in detail for several series of the bilayer high-T_c Cu oxides.[19-22)] In this letter, we address the question of what kind of low-energy spin excitation commonly underlies the doped high-T_c Cu oxides, independent of the number of CuO_2 planes in a unit cell, regardless of the appearence of the pseudo spin-gap behavior (iii). We shall focus on the crossover phenomena (ii) and the doping dependence (i).

We analyze newly measured $1/T_1$ of the planar ^{63}Cu for $La_{2-x}Sr_xCuO_4$ [$x(T_c) = 0.06$ (7 K), 0.15 (38 K) and 0.20 (30 K)],[9,23,24)] as well as the previously reported $1/T_1$ and $1/T_G$ for the other compounds,[10,14,21,22)] in respect to doping dependence of the high-T CW law in the SCR theory and the low-T deviation from it. We found the unconventional suppression of the low-frequency spin fluctuation $\chi(\boldsymbol{Q})$ near the metal-insulator (MI) Mott transition point. We propose the low-T suppression from the thermal $\chi(\boldsymbol{Q})$ as an anomaly inclusive of the pseudo spin-gap.

In Fig. 1, the experimental $1/T_1$ of the planar ^{63}Cu(A) are plotted versus temperature for $La_{2-x}Sr_xCuO_4$ ($x =$ 0.00, 0.06, 0.15 and 0.20).[8,9,23-25)] Here, the Cu(A) site represents a main part of the planar ^{63}Cu.[26)] At high T, $1/T_1$ appreciably changes with changing the Sr content,

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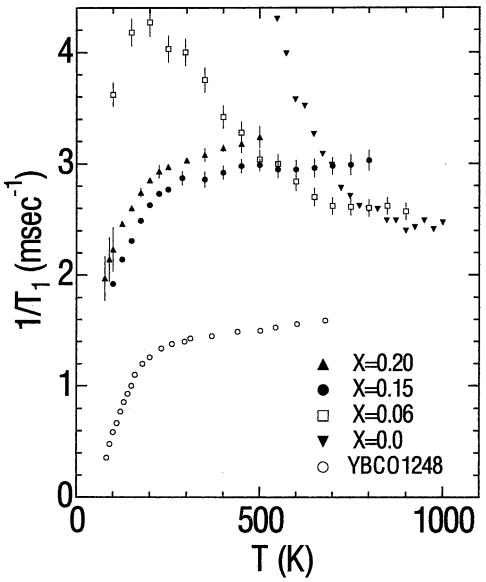


Fig. 1. The experimental $1/T_1$ of the planar ^{63}Cu(A) is plotted against T from refs. 9, 23–25. For comparison, $1/T_1$ for $YBa_2Cu_4O_8$ is also presented [T. Machi *et al.*: Physica C **173** (1991) 32 and H. Zimmermann *et al.*: Physica C **159** (1989) 681]. One should note that the magnitude of $1/T_1$ depends on the frequency-site A or B.

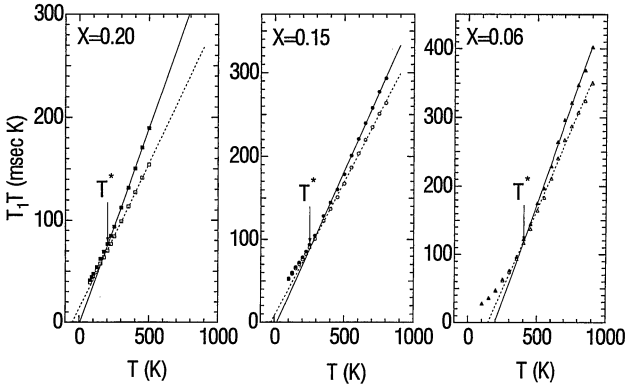


Fig. 2. The experimental T_1T (open symbols) and the antiferromagnetic part $(T_1T)_{AF}$ (closed symbols) of the planar ^{63}Cu(A) for $La_{2-x}Sr_xCuO_4$ are plotted against T. The solid and the short-dashed lines are the least-squares fits by a function $m(T+\Theta)$.

which is comparable with the doping dependent electrical resistivity.[27]

In Fig. 2, T_1T (open symbols) are plotted as a function of T.[9,23,24] Dynamical spin susceptibility $\chi(\boldsymbol{q}, \omega)$ of the CuO_2 plane is considered to consist of a strongly enhanced $\chi(\boldsymbol{Q})$ piled up on a $\boldsymbol{q}$-independent background component χ_{BG} $[\approx\chi(\boldsymbol{q}=\boldsymbol{0})]$.[28,29] We evaluate the antiferromagnetic parts $(T_1T)_{AF}$ (closed symbols in Fig. 2) weighted around the staggered mode $\boldsymbol{Q}$ using the planar oxygen nuclear spin-lattice relaxation rate $(1/^{17}T_1T)_{cc}$ as in refs. 7, 9, 29 and 30, i.e., $(1/^{63}T_1T)_{AF} = (1/^{63}T_1T) - k(1/^{17}T_1T)$. The factor k is a conversion factor defined in ref. 30. At present, the data of the relaxation rate $(1/^{17}T_1T)_{cc}$ and the Knight shift $^{17}K_{spin}$ are available only for the $x = 0.15$ below 250 K in ref. 31, where the relation $^{17}(K_{spin}T_1T) \approx$ constant is confirmed. Assuming this relation and the same constant value at all of the doping levels, we can evaluate $(1/^{17}T_1T)_{cc}$ using the bulk uniform $\chi(\boldsymbol{0})$.[32] Taking into account the hyperfine coupling constants,[9] we estimate $k(1/^{17}T_1T) = 0.37$, 0.75 and 1.90 $sec^{-1}K^{-1}$ for $x = 0.06$, 0.15 and 0.20, respectively.[9,24] The solid [or the short-dashed] lines for the respective materials are the least-squares fits to $(T_1T)_{AF}$ [or (T_1T)] by a function of $m(T+\Theta)$ with the free parameters m and Θ. The estimated m and Θ for $(T_1T)_{AF}$ and for T_1T are presented in Table I. The value of k depends on the form of the hyperfine coupling at the planar oxygen site.[33] Hence, the values of m and Θ estimated for $(T_1T)_{AF}$ depend on the amount of $k(1/^{17}T_1T)$. However, the tendency as a function of the Sr content does not strongly depend on it. In Fig. 2, $(1/T_1T)_{AF}$ [or $1/T_1T$] at high temperatures follows a CW law $1/m(T+\Theta)$. Low-T CW law below room temperature has been reported.[34] The anomalous sharp peaking of $1/T_1T$ just above T_c for the ^{63}Cu enriched materials has been observed by us.[35] Our primary concern is the high-T spin dynamics, because we study thermal spin fluctuations at high T.[3] In Fig. 2, $(1/T_1T)_{AF}$ at low T deviates from the CW law below a specific temperature T^*. We estimate the deviation temperature T^* to be 200, 250 and 400 K for the $x = 0.20$, 0.15 and 0.06, respectively. It is notable that the T^* increases with decreasing the carrier number toward the MI transition point.

First, we discuss the high-T CW law. Based on the SCR theory, Itoh presents an approximate expression for $(T_1T)_{AF}$ in terms of T_0 and T_A,[10]

$$(T_1T)_{AF} \propto (2\alpha_s T_A)[T+\Theta], \quad (1)$$

$$\Theta = 3.3 g_0 T_0, \quad (2)$$

with $g_0 \cong 0.0041 + 0.56 y_0$ for $|y_0| \ll 0.1$ at T ($>0.1T_0$). The parameters, α_s and y_0, are defined in ref. 3. Using eqs. (1) and (2), one can estimate T_A from the slope m of $(T_1T)_{AF}$ and T_0 from Θ.[3,10] In the phenomenological spin-fluctuation model,[28] the Weiss temperature Θ ($=T_x$ in ref. 28) and the spin-fluctuation energy T_0 ($=\Gamma$ in ref. 28) are independent free parameters to fit to the experimental data. On the other hand, in the SCR theory, eq. (2) is required from the fluctuation-dissipation theorem for the thermal spin fluctuation.[3] In the phenomenological model[28] (or the local-moment model),[29] the slope of m of $(T_1T)_{AF}$ represents the damping constant of the spin fluctuation Γ (the exchange frequency ω_{ex}).

In Table I, the sign (Θ) is negative for the present $La_{2-x}Sr_xCuO_4$, at least in the underdoping regime. Then, the ground state parameter g_0 is negative because the sign (Θ) = sign (g_0). This is consistent with the reported sign (Θ) for the underdoped compounds.[2,10] In the spin-fluctuatiom model on the SCR theory, Θ/m is proportional to g_0T_0/T_A $[\propto\chi(\boldsymbol{Q})\Gamma(\boldsymbol{Q})]$. If assuming the same $|g_0|$, the empirical ratio T_0/T_A for $1/T_1T$ in Table I seems to correlate with T_c. However, the values of $1/m$ ($\propto 1/T_A$; the value of $1/T_1$ at high temperatures) are different between the Cu AB sites.[24,35,36] Thus, the discussion on the magnitude of m for $La_{2-x}Sr_xCuO_4$ is limited. Theoretical development to include the local

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electronic state is desired.

In Fig. 3, T^* (closed symbols) and T_c (open symbols) are plotted against the hole number P_{sh} per a CuO_2 sheet ($Cu^{2+P_{sh}}$) on the ionic model;[37] $La_{2-x}Sr_xCuO_4$ (■, □),[8,9,23,24] $YBa_2Cu_3O_y$ (●, ○),[2,13,22] and $Y(Ba_{1-x}La_x)_2Cu_3O_y$ (▲, △),[21] $HgBa_2CuO_{4+\delta}$ (▼, ▽)[14] estimated from $(1/^{63}T_1T)_{AF}$ [or $(1/^{63}T_1T)$ or $(1/T_G)^2$]. The MI transition point is identified as $x \sim 0.04$ for $La_{2-x}Sr_xCuO_4$, $y \sim 6.42$ for $YBa_2Cu_3O_y$ and $(x, y) = (0.3 \sim 0.4, \sim 7.18)$ for $Y(Ba_{1-x}La_x)_2Cu_3O_y$. The insulating phase has not yet been confirmed for $HgBa_2CuO_{4+\delta}$. It is not clear why the monolayer $HgBa_2CuO_{4+\delta}$ exhibits a clear pseudo spin-gap behavior ($T_c \ll T_s \leq T^*$)[14] in the temperature dependence of $1/T_1T$ but not $La_{2-x}Sr_xCuO_4$ ($T_c \leq T_s \ll T^*$ where T_s *increases with* x).[35] Nevertheless, one can see the same aspect of the unusual suppression of the staggered spin fluctuation for both compounds. In Fig. 3, free from the detail of the definition of T^*,[38] the crossover temperature T^* increases with decreasing the hole number P_{sh}. At locations close to the MI transition boundary, the lower-T_c compound has a large T^* in the same series. This doping dependence of T^* could not be explained by the existing spin fluctuation theories as we discuss below.

SCR theory (a weak coupling theory): in the SCR theory, the thermal spin-spin correlator approaches the mean-field one at $T < T^*_{SCR}$ (a crossover temperature).[3,39] The T^*_{SCR} is proportional to the spin-fluctuation energy T_0; $T^*_{SCR} = 2\sqrt{2}y_0T_0$.[39] In the existing SCR theory,[3] a large Fermi surface is assumed and the disorder-to-order transition at $T = 0$ is supposed to correspond to the actual MI transition driven by varying the carrier density. Here, the disorder-to-order transition is the transition from a nearly antiferromagnetic metal to a weakly antiferromagnetic one. Reducing the carrier number may lead to reducing $y_0T_0 \to 0$, so that the T^*_{SCR} may decrease with decreasing P_{sh}.[3,40] The quantum critical state with $T^*_{SCR} = 0$ is also expected just at the critical boundary from the renormalization group theory.[41] The characteristic energy constant is expected to vanish at the critical boundary. However, the experimental T^*-vs-P_{sh} in Fig. 3 is not consistent with the theoretical T^*_{SCR}. We report the low-T deviation from the existing SCR theory as the anomaly, including the pseudo spin-gap, peculiar to the doped high-T_c Cu oxides near half-filling.

Mean-field t-J model: the change of sign of the Weiss temperature Θ can be reproduced by the mean-field calculation for the uniform RVB,[16] although the experimental CW law is not reproduced.[2] The spinon bandwidth ($\sim J$) is self-consistently determined via the holon order parameter, so that the system lies in the extreme critical region but the actual antiferromagnetic transition is prevented by the limited spinon bandwidth.[42] The spinon spin susceptibility is calculated based on a large spinon Fermi surface. The calculated T_1T decreases with decreasing T to a crossover temperature T^*_{tJ}, and T_1T below T^*_{tJ} shows a weak T-dependence. Here, we estimate a crossover temperature T^*_{tJ} from the calculated T_1T as an approximate function of the doping rate δ; $T^*_{tJ} \sim (4\delta - 0.25)J$, for $0.10 \leq \delta \leq 0.15$.[16] However, this doping dependence of T^*_{tJ} is not consistent with the experimental T^*.

A doped 2D quantum Heisenberg antiferromagnet (QHA): according to the renormalization group theory for QHA, the quantum-critical point realizes at $T = 0$ K as the order-to-disorder transition point.[43] If this point corresponds to the MI transition point, the characteristic energy scale should vanish. In this respect, an extrapolated value of the experimental T^* ($P_{sh} \to 0$) from the metallic side also seems to be inconsistent.

Within the above discussions, not only a weak coupling theory but also a strong coupling theory based on a large Fermi surface fail to reproduce the experimental doping dependence of T^*. The band structure must change significantly with doping, and the rigid-band picture breaks clearly.

It is interesting to note the similarity in the doping dependence between the experimental T^* and the Néel temperature T_N for the weakly antiferromagnetic 2D metal

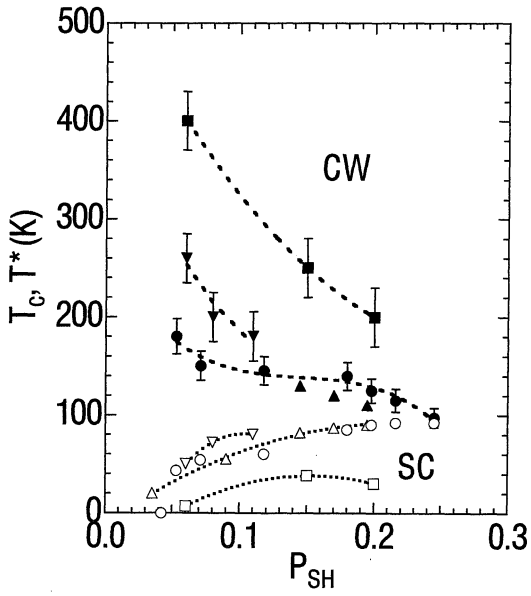


Fig. 3. The crossover temperature T^* (closed symbols) and the superconducting transition temperature T_c (open symbols) are plotted against the hole number P_{sh}; $La_{2-x}Sr_xCuO_4$ [x = 0.06, 0.15, 0.20] (■, □), $YBa_2Cu_3O_y$ [y = 6.42, 6.45, 6.50, 6.63, 6.80, 6.85, 6.90, 6.98] (●, ○), $Y(Ba_{1-x}La_x)_2Cu_3O_y$ [(x, y) = (0.1, 6.98), (0.05, 6.94), (0, 6.89)] (▲, △) and $HgBa_2CuO_{4+\delta}$ [δ = 0.03, 0.04 and 0.055] (▼, ▽). The dashed curves are guides for the eye. CW=Curie-Weiss law and SC=superconducting state.

Table I. The superconducting transition temperature T_c, the results of least-squares fits to $(1/^{63}T_1T)_{AF}$ [or $(1/^{63}T_1T)$] by a function of $1/m(T + \Theta)$, and the crossover temperature T^* for $La_{2-x}Sr_xCuO_4$.

$La_{2-x}Sr_xCuO_4$	$x = 0.06$	0.15	0.20
T_c (K)	7	38	30
Θ (K)	−200 (−150)	−20 (25)	−5 (30)
$1/m$ (msec^{-1})	1.8 (2.1)	2.6 (3.1)	2.6 (3.6)
Θ/m (sec^{-1}K)	−0.36 (−0.32)	−0.05 (0.08)	−0.01 (0.11)
T^* (K)	∼400	∼250	∼200

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($y_0 < 0$ state in ref. 3).[44] Below T_N, the partial lack of a Fermi surface may be expected, due to formation of the anisotropic SDW gap. Also, the similarlity between the T^* and the mean-field T_{RVB} for the singlet RVB formation has been reported.[11,14,21] No phase transition, but the crossover phenomena takes place at the actual T^*. These similarities are only suggestive for clarifying the electronic states.

In conclusion, we observed an unusual suppression of the staggered spin fluctuation below a crossover temperature T^* in the doped high-T_c Cu oxides. We found that the increase of the crossover temperature T^* near the critical boundary cannot be understood within the existing spin-fluctuation theories where the characteristic energy constant is reduced to zero near the MI transition point.

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38) In a monolayer system, we have $(1/T_1T)_{AF} \sim \chi_{/\!/}$ and $(1/T_G)^2 \sim (\chi_{/\!/})^2$, where $\chi_{/\!/}$ is the in-plane susceptibility. According to Kishine in ref. 18, in a bilayer coupled system the expressions should be modified by $(1/T_1T)_{AF} \sim [\chi_{/\!/} + \Delta\chi_{/\!/} + \Delta\chi_\perp]$ and $(1/T_G)^2 \sim [(\chi_{/\!/} + \Delta\chi_{/\!/})^2 + (\Delta\chi_\perp)^2]$. Here, $\Delta\chi_{/\!/}$ and $\Delta\chi_\perp$ are the diagonal and off-diagonal counterparts of the inter-layer exchange scattering process, respectively. These extra terms of $\Delta\chi_{/\!/}$ and $\Delta\chi_\perp$ are rapidly suppressed above T_s. Therefore, we can expect that the present analysis does not change even in the bilayer coupled system.
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